Warm plasma model dispersion relation

Virtual in class exercise October 5 & 7 2026

Part 1

Recall that the relation for $\boldsymbol{J}_{s,1}$ in terms of $\boldsymbol{E}_1$ can be solved to identify $\overleftrightarrow{\sigma}$, hence $\overleftrightarrow{\chi}$. Without making further approximations, the goal is to find and solve the characteristic equation with this full warm plasma susceptibility. To do so, you may use your choice of language.

Step 1: Derive the eigenvalue problem analytically based on the above instructions.

Step 2: Solve the eigenvalue problem e.g. using Mathematica.

Part 2

Step 3: Plot the results for the following conditions:

- Overdense ($\omega_{p,e} > \omega_{c,e}$) with $\theta > 0$ and $T_{i,0} = 0$.
- Overdense with $\theta < \pi/2$ and $T_{i,0} = 0$.
- Overdense with $\theta = \pi/4$ and $T_{i,0} \neq 0$.
- Underdense with $\theta = \pi/4$ and $T_{i,0} \neq 0$.

General notes:

1. Show $\omega(k)$ on a $\log_{10} - \log_{10}$ scale.
2. If $\omega(k) \propto k^\alpha$ then $\log_{10}(\omega) = \log_{10}(\text{const.}) + \alpha \log_{10}(k)$.
   - The slope is the α and, for all non-dispersive waves, is equal to unity.
   - For non-dispersive waves, the phase speed is the constant.
3. Frequency (vertical axis) is normalized by $\omega_{c,i}$ and wave number (horizontal axis) is normalized by δ_i^{-1} where $\delta_i = c/\omega_{p,i}$ is the ion skin depth.
   - For $\omega_{p,i}^2 \gg \omega_{c,i}^2$, $V_A = c\omega_{c,i}/\omega_{p,i} = \delta_i \omega_{c,i}$.
   - Ion responses are unmagnetized for $k\rho_s > 1$ where $\rho_s = c_s/\omega_{c,i}$ is the sound gryo-radius and

     $$c_s = \sqrt{(k_B\gamma_i T_i + k_B\gamma_e T_e)/m_i} \tag{1}$$

     is the ion acoustic speed ($q_i = e$) such that $\rho_s/\delta_i = c_s/V_A$.
4. Transitions: Solutions cross in $\omega(k)$ only when there is no coupling in the physics of the two polarizations. Crossings occur for principal modes. Thermal effects contribute to mixed physics responses that transition solutions from one behavior to another, avoiding some crossings for $0 < \theta \leq \pi/2$.
5. Waves with v_p (or v_ϕ in the plots) $\approx c_s = \sqrt{\gamma_s k_B T_s/m_s}$ also have $v_p \approx v_{th,s}$, where the fluid modeling is inaccurate. However, for $T_e \gg T_i$, $v_p = c_e$ occurs with $c_i \ll v_p \ll c_s$, which satisfies the fluid approximation.

Step 4: For each case, explain in detail the physical interpretation and consequences of the warm plasma model e.g. how does it impact the wave behavior compared to the cold plasma model?